MALACKATHON
INFORMATICA UMA

UNIVERSIDAD DE MÁLAGA | uma.es

III MALACKATHON 2026

INDRA GROUP

CHALLENGE TITLE: Detection and characterization of a wildfire using satellite imagery

Are you interested in INDRA Space?

We invite you to take part in the 3rd Malackathon and tackle the challenge set by INDRA Space, designed to apply your skills to a real-world problem.

Wildfires are among the natural phenomena with the greatest environmental, economic, and social impact. Up-to-date information on their extent and development is essential to support monitoring, damage assessment, and decision-making.

Earth observation satellites can periodically capture images of large areas of land. In addition to visible-color imagery, onboard sensors can observe different regions of the electromagnetic spectrum, providing information that is not directly perceptible to the human eye and may be especially useful for analyzing phenomena such as wildfires.

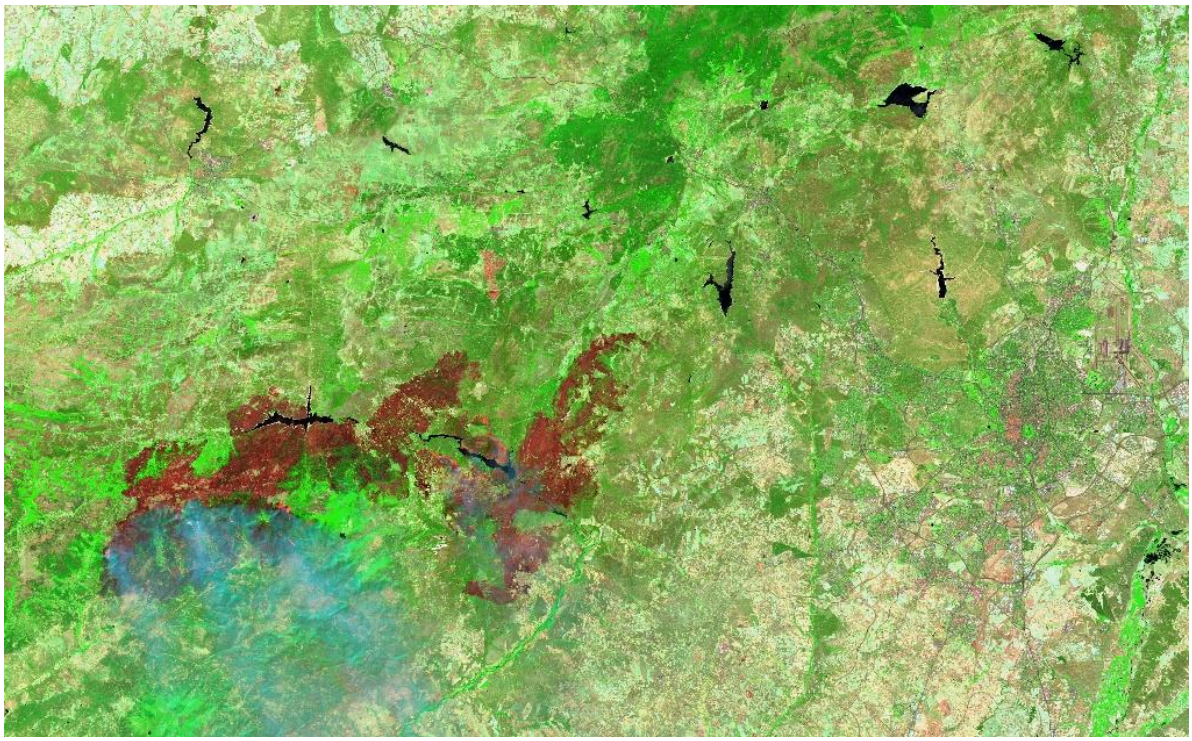
For this challenge, two images captured by the Sentinel-2 satellite of the European Copernicus program will be provided. They show the same area at the same acquisition time during wildfire events in the following locations:

Avila-Madrid. The images represent the same scene using two different composites: a visible-color image (True Color) and a composite incorporating short-wave infrared (SWIR) information.

Niebla (Huelva). The images represent the same scene using three different composites: a visible-color image (True Color), a composite incorporating short-wave infrared (SWIR) information, and an image highlighting the wildfire front (Wildfire).



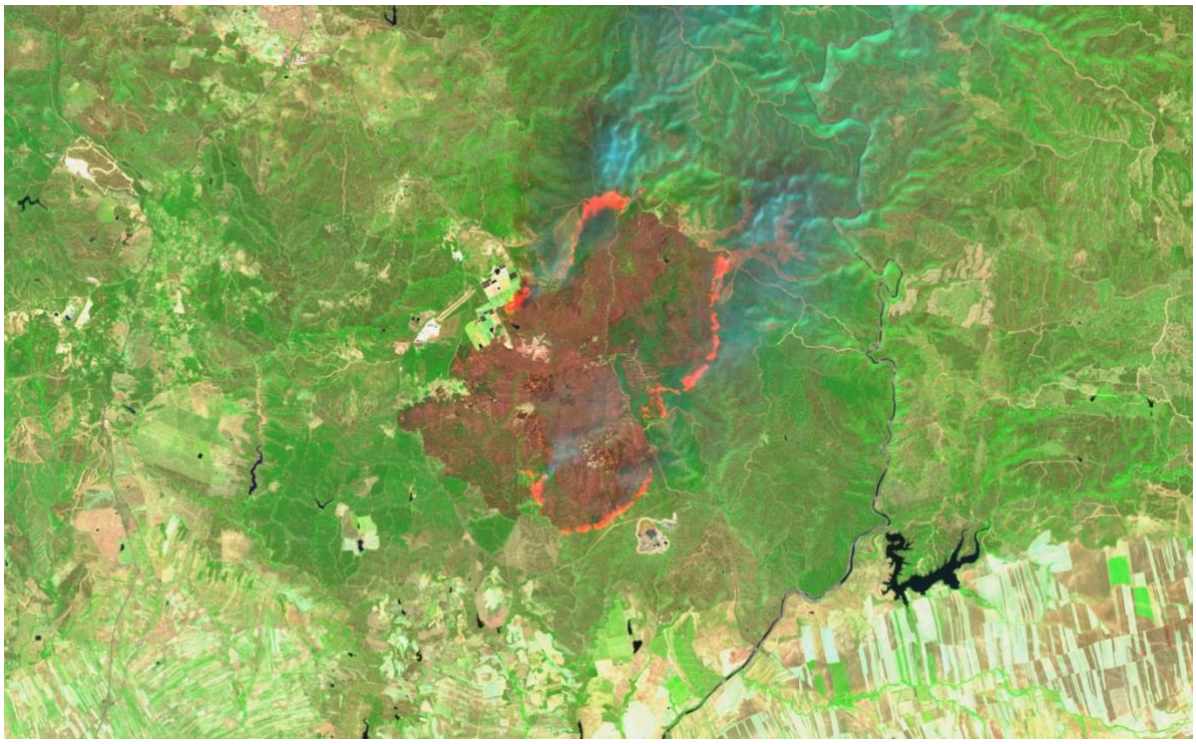
(Avila-Madrid True Color Image)



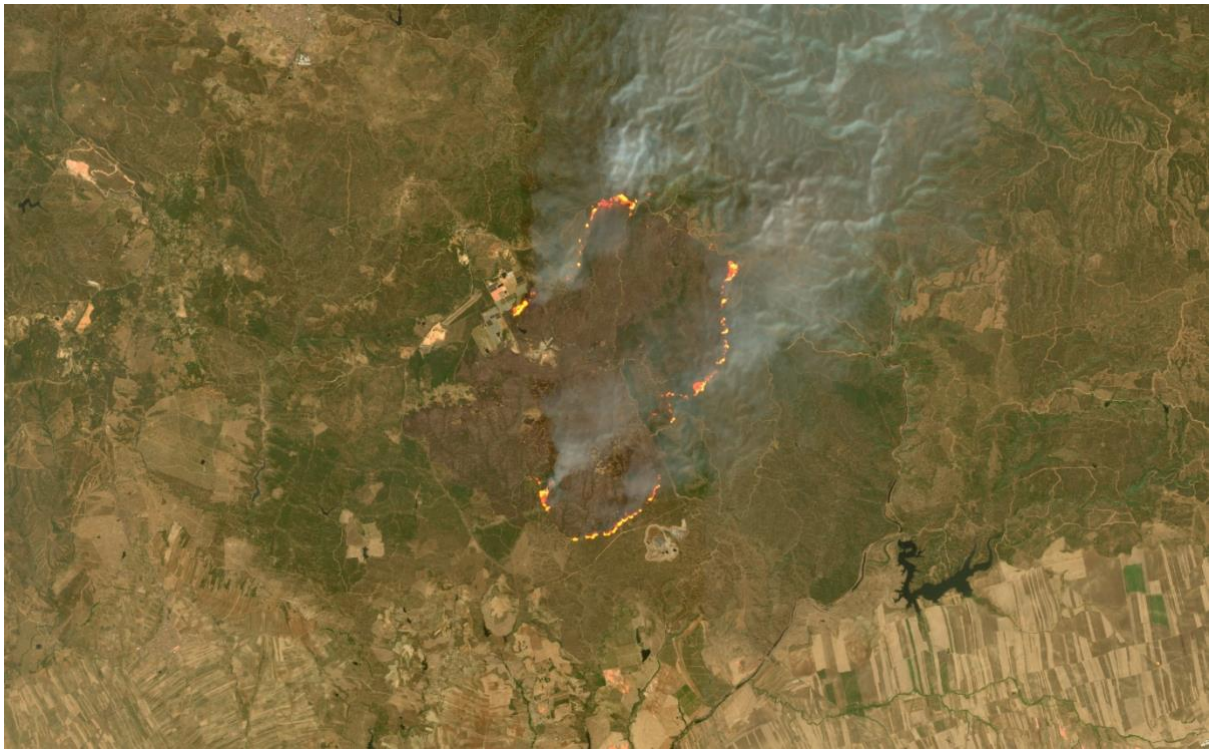
(Avila-Madrid SWIR Image)



(Niebla True Color Image)



(Niebla SWIR Image)



(Niebla Wildfire Image)

The Malackathon includes two types of challenges: basic and advanced. The advanced challenges provide access to Friday's final. Each advanced challenge will be assessed by mentors from the sponsors of the corresponding prize. To be assessed for any advanced challenge, you must have fully completed all the basic challenges.

THE CHALLENGE

The objective is to develop, within 48 hours, a software solution capable of automatically analyzing the provided images and extracting relevant information about the wildfire.
The solution must address two main objectives:

Basic. Delineation of the area affected by the wildfire. The system must identify the region showing evidence of wildfire impact and estimate its perimeter. The result must be displayed graphically over one of the original images, clearly distinguishing the detected area from the rest of the territory.

Advanced. Detection of possible active-fire areas. The system must identify regions or points in the image showing evidence consistent with fire activity at the time the satellite captured the image. These areas must be clearly distinguished from affected surfaces with no apparent activity.

The solution must take into account other elements present in the images—such as vegetation, bare soil, bodies of water, urban areas, clouds, or smoke—that may hinder detection and produce false positives.

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FREEDOM OF APPROACH

No specific algorithm, technology, or methodology is prescribed. Participants may use digital image processing, computer vision, statistical analysis, machine learning, artificial intelligence, or any combination of these approaches.

Particular value will be placed on the teams' ability to investigate the problem, correctly interpret the available information, and justify their technical decisions, as well as on the quality of the result obtained.

At a minimum, the solution must produce a visual representation that clearly identifies the estimated perimeter of the affected area and the areas detected as possible active fire hotspots. Teams may also include segmentation masks, coordinates, estimated surface areas, confidence levels, statistics, interactive maps, or any other outputs they consider useful.

EVALUATION

The evaluation will consider the quality of the detection of the affected area and active-fire hotspots, the reduction of false positives, the robustness and level of automation of the solution, the technical quality of the developed software, the clarity of the results presentation, and the originality of the approach.

In addition to the images used during development, the organizers may use additional images not previously known to participants to assess the solutions' ability to generalize and to prevent the algorithms from being tailored exclusively to a specific scene.

The ultimate goal is not merely to achieve good image segmentation, but to build, within 48 hours, a software prototype capable of transforming satellite observation data into useful, easily interpretable information about the extent and activity of a wildfire.

PRIZE NAME

Teams will consist of 4 or 5 people

COMPANY:

Prize: Tablets

Title: Rewarding solutions that put users at the center by creating truly accessible digital tools for everyone.

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